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Heterogeneous Integration Technologies for Artificial Intelligence Applications

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ABSTRACT The rapid advancement of artificial intelligence (AI) has been enabled by semiconductor-based electronics. However, the conventional methods of transistor scaling are not enough to meet the exponential demand for computing power driven by AI. This has led to a technological shift toward system-level scaling approaches, such as heterogeneous integration (HI). HI is becoming increasingly implemented in many AI accelerator products due to its potential to enhance overall system performance while also reducing electrical interconnect delays and energy consumption, which are critical for supporting data-intensive AI workloads. In this review, we introduce current and emerging HI technologies and their potential for high-performance systems. We then survey recent industrial and research progress in 3-D HI technologies that enable high bandwidth systems and finally present the emergence of glass core packaging for high-performance AI chip packages.

INDEX TERMS Artificial intelligence (AI), glass packaging, heterogeneous integration (HI), packaging.

I. INTRODUCTION

The introduction of ChatGPT in 2022 sparked an exponential growth in artificial intelligence (AI) and high-performance computing (HPC) applications, making AI increasingly more important to daily life. Large AI models excel at handling complex tasks; however, they require large training datasets and sizeable computing systems. These large compute workloads lead to larger die sizes and higher power densities, making it more difficult to design energy-efficient architectures [1], [2], [3], [4]. However, demand for compute continues to grow even as the traditional rate of scaling slows down [5], [6], [7].

Thus, the heterogeneous integration (HI) of chiplets is essential to achieve high system throughput (tera-operations-per-sec, or TOPS) and energy efficiency (TOPS/W) to meet the growing computational demands [8], [9], [10]. By splitting a system-on-chip (SoC) into multiple chiplets and integrating them into a single package, the system's design flexibility, functionality, bandwidth, throughput, and latency can be significantly improved. This can be achieved by bringing chiplets closer together laterally, vertically, or even both ways, allowing more memory or logic to be integrated within a single package [11], [12], [13]. Furthermore, reducing the die sizes and performing known good die (KGD) testing before packaging can enable a higher level of control in die performance, resulting in yield improvement and reduced overall costs [14], [15], [16].

HI is a potential solution to realize high-performance systems that are dedicated to training large generative AI models. By integrating dice such as high bandwidth memory (HBM), central processing units (CPUs), and graphics processing units (GPUs) into a single package, the throughput, latency, and energy efficiency are significantly improved and overcome the limitations of conventional 2-D monolithic chip designs [17], [18]. Today, semiconductor companies such as Nvidia, Intel, and AMD have leveraged HI technologies in their own products to run real-time generative AI models and train large language models (LLMs) with over billions of parameters. In this review, we first introduce current and emerging HI technologies and discuss their advantages and current limitations. Then we survey recent commercial deployments of HI architectures that are designed for

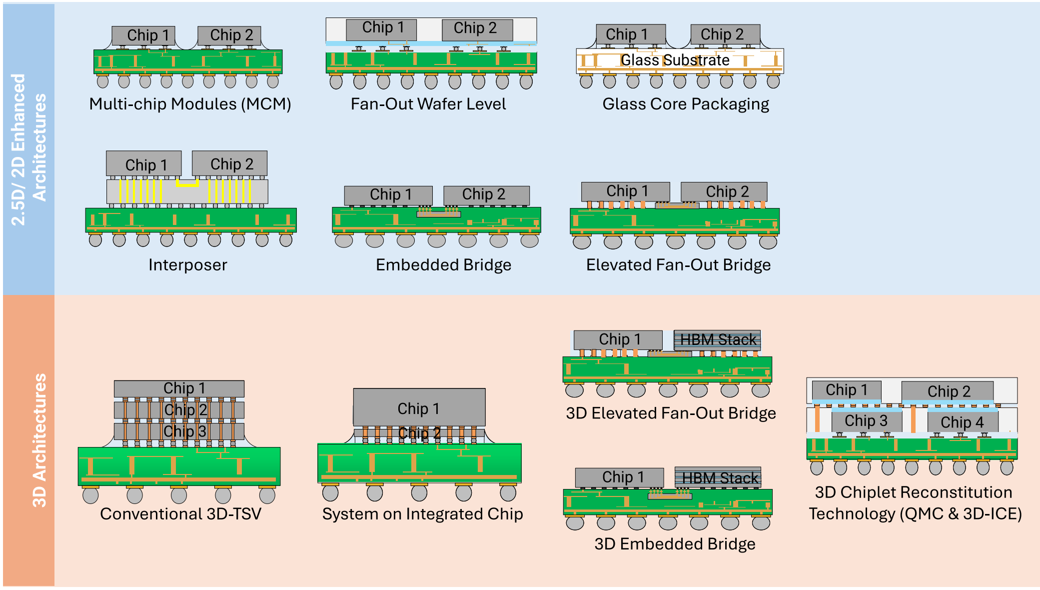

FIGURE 1. Landscape overview of current HI technologies (adapted from [22]).

TABLE 1. Summary of Current HI technologies.

Integration Platform	**Multi Chip Module (MCM) [26]–[29]**	**Fan-out Wafer Level Packaging (FOWLP) [40]–[45]**	**Interposer Technology [46], [47]**	**Bridge-based Technology [48]–[53]**	**3D Integration [11], [69]–[72]**
Examples	Flip Chip	ASE FOCoS, TSMC InFO, Amkor SWIFT etc.	Silicon, Organic, Glass etc.	EMIB, EFB, DBHI, GT HIST etc.	TSV-based, Hybrid Bonding-based (SoIC, QMC)
Off-chip I/Os	C4 bumps	C4 bumps, C2 bumps, Multilayer RDL	Microbumps, Multilayer Metal Stack	Microbumps, Multilayer Metal Stack, Compressible MicroInterconnect (CMI)	TSVs, Hybrid Bonds etc.
I/O Pitch	$>$130 μm	$>$130 μm, 20 - 44 μm	35 - 50 μm	45 - 55 μm	TSV: $>$10 μm Hybrid Bond: $<$ 10 μm
Link Wire Length (mm)	40 - 50 mm (Organic)	$>$ 100 μm to mm range (based on D2D spacing)	$>$ 430 μm to mm range (based on D2D spacing)	$>$ 1 mm	NA (Depends on interconnect height)
Line/Spacing (μm)	$>$10/10 μm	0.8/0.8 - 2/2 μm	$>$ 0.4/0.4 μm	$>$ 1/1 μm	NA
Parasitic Capacitance	NA	NA	0.4 - 1.61 nF	118 fF/mm	TSV: 31.5 fF Hybrid Bond: $<$ 1 fF
Example D2D Interface	XSR, UCIe	AIB, UCIe	XSR,AIB, UCIe etc.	AIB, UCIe etc.	AIB, UIB, UCIe etc.
Link Latency	XSR: 5 ns UCIe: $<$ 2 ns	UCIe: $<$ 2 ns	NA	UCIe: $<$ 2 ns	TSV: $<$ 100 ps
Power Consumption	$<$ 0.5 pJ/bit	$<$ 0.5 pJ/bit	$<$ 1 pJ/bit	0.17 - 0.7 pJ/bit	0.17 pJ/bit
Bandwidth	XSR: 800 Gbps/mm die edge UCIe: 188 - 1350 Gbps	UCIe: 188 - 1350 Gbps	AIB 1.0: 1.25 Gbps/mm (Shoreline BW) UCIe: 188 - 1350 Gbps	$>$ 3 Gbps	307 Gbps (Wafer scale hybrid bond)

high-compute AI workloads from semiconductor companies such as Cerebras, Nvidia, AMD, Intel, and Tesla. Finally, we survey recent advances in glass core packaging and evaluate their advantages and limitations.

II. CURRENT TRENDS OF HI TECHNOLOGIES

Partitioning an SoC into chiplets has largely been motivated by increased system functionality and lower manufacturing costs [19], [20], [21]. To enhance the performance of these chiplet-based systems, there have been multiple innovations in multidie HI architectures. We classify multidie architectures as 2-D, 2.5-D, or 3-D as defined by the IEEE Electronics Packaging Society (EPS) HI Roadmap [22] and provide an overview in Fig. 1. Table 1 gives a summary of the current HI technologies.

A. MCM ARCHITECTURE

Multichip modules (MCMs) are one of the earliest multidie 2-D architectures where chiplets are placed together laterally on an organic substrate to reduce wire length and increase package bandwidth to increase system performance and design flexibility [24], [25], [26], [27], [28], [29]. This is one of the simplest integration technologies; however, MCMs can have limitations in interconnect density due to the use of conventional organic substrates and coarse solder-based bonding technologies [30], [31]. These solder-based interconnects, such as C4 bumps, are challenging to scale down to finer pitches due to adjacent interconnects shorting during the bonding process [32], [33], which limits the system performance. For large AI systems, low latency and efficient memory access are required; however, scaling MCMs to larger systems is difficult due to the limited interconnection, which can become a bottleneck [30].

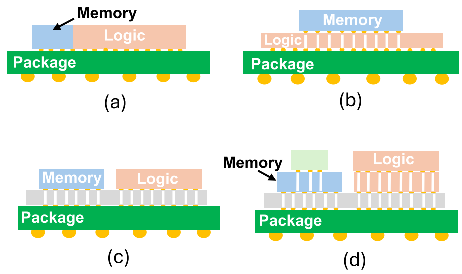

FIGURE 2. (a) Conventional 2-D SoC partitioned in (b) 3D-TSV-based architecture, (c) interposer architecture, and (d) combined architecture (adapted from [11]).

B. INTERPOSER ARCHITECTURES

These challenges have led to 2.5-D architectures that use substrates such as glass, silicon interposers, or localized silicon bridges to improve the lateral interconnect density [34], [35], [36]. Fine-pitch microbumps and through-silicon-via (TSV) technologies enable increased interconnection density for chiplets stacked on a glass or silicon interposer [37], [38], [39]. However, as computing demand grows, it can be expensive to scale interposers to large-scale AI systems [48], [49], [50], [51]. Therefore, bridge-based architectures such as Intel's embedded multidie interconnect bridge (EMIB), use localized silicon embedded in the package substrate with multiple routing layers for finer wiring pitches [48], [49]. Interdice signals are in the localized silicon bridges and power/ground interconnects and other signals are in the organic package to eliminate the need for TSVs and simplify the assembly process [48], [49]. Similar to EMIB, elevated fan-out bridge (EFB) uses a localized silicon bridge for increased interdice interconnect density with the bridge above the package substrate [51], [52]. This method can further reduce the assembly cost and complexity. Bridge-based technologies are promising for large-scale AI systems because of increased design functionality, lower design complexity, and easier thermal management compared with 3-D HI; however, conventional interconnect technologies such as microbumps can limit their system performance [54], [55]. This has led to new bonding technologies such as copper-to-copper bonding as potential solutions to overcome this limitation [56], [57]. Georgia Tech has proposed low-loss interconnects, known as heterogeneous interconnect stitching technology (HIST), as a potential solution for high-density 2.5-D integration [58].

C. WAFER-LEVEL PACKAGING

Wafer-level packaging (WLP) techniques are of great interest for advanced chiplet-based architectures as they enable high interconnection density, reduced interconnect delay, and increased bandwidth [40], [41], [42], [43]. High integration density can be achieved by fanning out chip I/O signals, instead of using traditional interconnects such as wirebonds or C4 bumps, making WLP suitable for high-performance systems [17], [44], [59]. In conventional WLP, the KGDs are encapsulated in an epoxy mold compound (EMC) to form a reconstituted wafer [60], [61]. However, EMC can cause manufacturing issues due to mismatch in the coefficient of thermal expansion (CTE) between the EMC and the dies which can result in warpage and die shifts/misalignments, and the low thermal conductivity of the material makes power dissipation difficult for high-powered systems [62], [63]. Therefore, alternative materials have been proposed to embed/encapsulate chiplets [64], [65], [66], [67].

D. THREE-DIMENSIONAL ARCHITECTURES

Three-dimensional HI technologies are a promising approach to meet the computing demands of AI systems [68], [69]. Using TSV and fine pitch interconnect technologies, such as microbumps or hybrid bonding, 3-D stacking can enable high bandwidth and low latency systems [11], [24], [70], [71], [72]. Many semiconductor companies have developed their own 3-D architectures, including Intel's Foveros [73], Samsung's X-Cube [74], and AMD's 3-D V-Cache product that uses TSMC's system on integrated chip (SoIC) technology [11]. SoIC technology partitions an SoC into multiple chiplets that can be reintegrated into various 3-D configurations. This allows the flexibility to integrate passive and active chips at different technology nodes, materials, and chip sizes (see Fig. 2) to support memory bandwidths over 20 Tbps [11]. Hybrid bonds have drastically increased bond density by 16× compared with conventional 3-D IC microbumps and reduced electrical parasitics such as IR drop and lowering energy consumption per bit [11]. In addition to finer interconnect pitch, SoIC technology has a higher metal routing density and thinner bonding layer, which can improve thermal performance. However, the technology faces similar challenges to conventional 3-D ICs. Scaling down hybrid bond pitch becomes increasingly more difficult due to the strict surface cleanliness and chemical mechanical polishing (CMP) requirements [75]. It is important to note that the 3-D system bandwidth is determined by the total number of stacks and size of the bottom die [55]. Although increasing the number of dice in a 3-D stack is desirable to increase memory bandwidth or computing power, assembly complexity and costs can increase significantly. In addition, thermal dissipation and

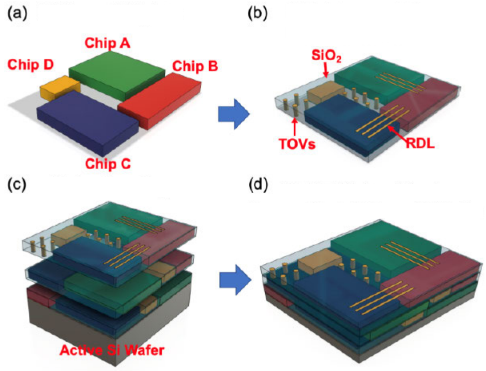

FIGURE 3. Overview of 3D-ICE technology [66]. (a) Different chiplets. (b) SiO_2 encapsulation reconstituted tier. (c) Wafer bonding. (d) 3D ICE wafer stack: reconstituted tier + Si wafer.

mechanical stability become more difficult. Liquid cooling has been proposed as a potential solution to help with heat dissipation [76]; however, this area is out of the scope of this article.

Other 3-D architectures have recently emerged using WLP technologies. Li and Bakir [66] proposed a wafer-scale chiplet reconstitution technology referred to as 3-D integrated chiplet encapsulation (3D-ICE), where multiple chiplets are encapsulated in low-temperature SiO_2 to form a reconstituted-SiO_2 tier, as shown in Fig. 3. This SiO_2 tier could then be postprocessed to enable high-density 3-D HI. Similarly, Intel presented the quasi-monolithic chip (QMC) as a new 3-D HI architecture where chiplets are also encapsulated in an ultrathick silicon dioxide layer [70]. SiO_2 as an encapsulant material offers several benefits. It can facilitate high-speed signaling due to its low-loss properties and there are essentially no die shifts or misalignments since curing is not required and it is compatible with the existing CMOS fabrication processes, thereby blurring the lines between package processing and device processing.

Although SiO_2 facilitates excellent electrical performance, the material has a low thermal conductivity, which can result in poor thermal performance. Therefore, Victor et al. [77] proposed a chiplet reconstitution process with an integrated heat spreader. The 30-μm-thick passive chiplets were encapsulated in 15-μm-thick ICP-PECVD SiO_2. Oxide deposited on top of the chiplets was etched away, and then 36 μm of copper was electroplated on the chiplets. The monolithic copper heat spreader helped reduce the maximum junction temperature of the chiplet tier, thereby addressing the electrical and thermal performance tradeoffs that most FOWLP solutions suffer from.

III. HI TRENDS FOR AI

A. CURRENT LANDSCAPE OF HI PRODUCTS

The rapid growth of AI has driven multiple commercial deployments of HI architectures specifically designed to accelerate the largest AI workloads. In this section, we survey recently reported industry products and summarize their specifications in Table 2.

In 2024, Cerebras introduced WSE-3, a wafer-scale AI accelerator that is twice as fast as WSE-2, and designed to train models 10× larger than GPT-4 and Claude [78]. Interestingly, Cerebras uses traditional device scaling and wafer-scale integration to go beyond Moore's law. With TSMC's 5-nm technology, four trillion transistors are fabricated on a single wafer with a chip size that is appropriately 57× larger compared with a GPU. However, the compute and memory components are decoupled to enable memory capacity scaling, so a single WSE-3 system is capable of storing and training models with 24 trillion parameters much more efficiently than a cluster made up of 10 000 GPUs.

Compared with Cerebras, other semiconductor companies are using advanced packaging technologies to design large-scale AI systems. Nvidia announced its GB200 Grace Blackwell chip that is made up of two Blackwell GPUs and one Grace CPU [79]. This chip is designed for LLMs with more than ten trillion parameters and 384 GB of off-chip memory, with a total device power of 2700 W. To achieve this, Nvidia uses TSMC's chip-on-wafer-on-substrate (CoWoS)-L packaging technology [79]. This packaging technology uses local silicon interconnect (LSI) chips and a reconstituted interposer to achieve a large integration area, bandwidth, and low latency high-performance system [37].

AMD uses a chiplet approach for its MI300X package and combines interposer technology and 3-D stacking to achieve high performance and memory bandwidth [80]. The MI300X is made up of multiple GPU chiplets, I/O dies, and 192 GB of HBM and has a total device power of 750 W. CPU complex dice (CCDs) and accelerator complex dice (XCDs) are 3-D stacked on I/O dice (IODs) for low signal latency. Finally, the 3-D stacks and HBM dice are integrated using a large silicon interposer to achieve a high-performance system [81].

Intel's Gaudi-3 Accelerator product uses their embedded bridge chip technology to integrate two Intel compute dies with 128 GB of HBM to enhance large-scale AI systems [82], [83]. Similar to other bridge-based interposer technologies, EMIB allows Intel to increase design functionality and reduce assembly cost [48], [49]. Although the Gaudi-3 Accelerator is not as powerful as Nvidia's H100, it is a cost-effective high-performance system.

Finally, Tesla has entered the AI market with Dojo, a chip that is optimized for large neural network training [84]. With a total device power of 400 W, much lower compared with competitors, Dojo is specifically designed for real-time data processing of driving situations. Tesla is using TSMC's integrated fan-out system-on-wafer (InFo-SoW) technology to achieve a high-density, low latency system [84], [85]. In summary, as AI models continue to grow in size and complexity, there has been a technological shift toward HI and emerging HI technologies.

B. D2D INTERFACES AND COMMUNICATION PROTOCOLS

As the number of chiplets is increasing within a single system, die-to-die (D2D) interfaces are becoming increasingly more

TABLE 2. Summary of State-of-the-Art AI hardware accelerators.

Parameters	Cerebras [78]	Nvidia [79]	AMD [80]	Intel [82]	Tesla [84]
Product name	Wafer-Scale Engine 3 (WSE-3)	GB200 Superchip	MI 300X	Gaudi 3	Dojo D1 chip
No. of transistors	4 Trillion	NA	158 Billion	NA	50 Billion
Cores	900,000	72(CPU Cores)	1216 (Matrix Cores)	64 (Tensor Processor Cores)	354 (Training Nodes)
Package Size	46,225 mm^2	NA	NA	NA	645 mm^2
AI compute (BF16)	125 PFLOPS	10 PFLOPS	1.3 PFLOPS	1.8 PFLOPS	365 TFLOPS
On-chip memory	44 GB	114 MB	256 MB	96 MB	440 MB
Off-chip memory	12 TB - 1.2 PB	384 GB (HBM 3e)	192 GB (HBM 3)	128 GB (HBM 2e)	NA
Memory bandwidth	21 PBps	16 TBps	5.3 TBps	3.7 TBps	2 TBps/edge
Interface	NA	NVLink 5	PCIe Gen 5 x16	PCIe Gen 5 x16	PCIe Gen 4 x16
Total Device Power(TDP)	23,000 W	2700 W	750 W	900 W (Passive cooling) 1.2 kW (Liquid cooling)	400 W
Technology node	TSMC 5 nm	TSMC 4NP	TSMC 5 nm	TSMC 5 nm	TSMC 5 nm
Packaging Technology	Wafer-Level	CoWoS-L	CoWoS and SoIC	EMIB	InFo-SoW

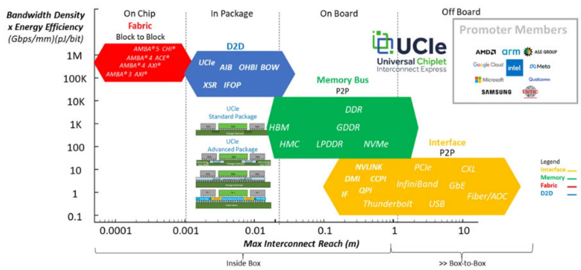

FIGURE 4. Standard protocols of heterogeneous computing [91].

(a) (b)

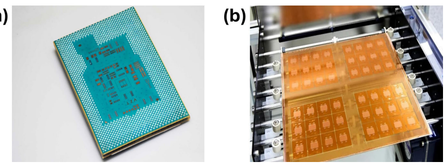

FIGURE 5. (a) Intel's assembled glass substrate test chip [92]. (b) Absolics' panel-level-packaging glass substrate [93].

critical for data movement between various components [86], [87], [89], [90]. AMD's Infinity Fabric [86] and Intel's Advanced Interface Bus (AIB) are D2D interfaces [87] that are used in their AI accelerator products to minimize latency and maximize bandwidth. However, as systems become more diverse with dice supplied by various vendors, the Universal Chiplet Interconnect Express (UCIe) protocol has started to rise as a common industry standard [88], [89], [90]. Standard D2D protocol is crucial for design flexibility and scalability, especially for large-scale AI and HPC systems, as well as networking systems. A summary of the different standard protocols for heterogeneous computing is shown in Fig. 4.

IV. GLASS PACKAGING

A. EMERGENCE OF GLASS-CORE SUBSTRATE PACKAGING

AI applications typically require larger interposers and very high-density interconnects to enable high bandwidth. These stringent requirements combined with reliability and performance demand advanced packaging technologies to be developed and implemented to construct large packages. With the needs for more advanced packaging techniques suitable for AI and HPC applications, using glass as the core substrate has gained tremendous attention lately due to its numerous advantages [67], [94]. Intel recently showcased their first glass substrate test chip and announced their trajectory toward glass packaging to meet the demands for more powerful compute. [Fig. 5(a)] Absolics Inc., a subsidiary of SKC in South Korea, has also begun preparing for small volume manufacturing (SVM) of their glass substrates [Fig. 5(b)], aimed for hyperscalers such as Amazon, Meta, and Microsoft as their prospective customers.

B. ADVANTAGES OF GLASS-CORE PACKAGES

Glass-based interposers enhance the bandwidth capabilities of semiconductor packages used in AI applications by improving signal integrity, supporting high-density interconnections, integrating optical communication, optimizing thermal management, and ensuring reliability and scalability. These characteristics make glass interposers essential components for achieving HPC and enabling advanced AI functionalities.

Glass, with its smooth surface/very low surface roughness, enables scaling of fine lines and spaces which is critical for achieving very high-density interconnects [95]. Furthermore, glass' surface structure consisting of Si–O linkages helps with the adhesion of a variety of polymer materials for use as dielectric and photosensitive resins. Combining glass' low dielectric constant together with the utilization of low dielectric constant buildup layers for multilayer interposer structures, the latency of the system can be reduced significantly. This property plays a crucial role in minimizing signal propagation delays and reducing crosstalk among adjacent interconnects, particularly beneficial for high-speed electronics and copackaged optics. In addition, glass substrates lower the capacitance between interconnects, enabling faster signal transmission and enhancing overall system performance [96]. In critical applications such as data centers, telecommunications, and HPC, where speed is paramount, the adoption of glass substrates can greatly improve system efficiency and increase data throughput.

Furthermore, the low dielectric constant of glass also supports superior impedance control, crucial for maintaining signal integrity throughout the circuit. This feature is especially advantageous in RF applications, where precise impedance matching is critical to optimize power transfer and minimize signal loss. Glass substrates ensure consistent electrical properties across the substrate surface, enabling the design and production of high-frequency circuits with improved reliability and performance [97].

Moreover, the exceptional dimensional stability of glass compared with organic packages helps improve the layer-to-layer accuracy which is key to enabling very high interconnect density in a multilayer glass interposer [98]. This helps reduce not only the pad size but also scaling of fine lines and traces to <1 μm, which increases the number of IOs in each redistribution layers (RDLs) in a multilayer interposer [99], [100]. In addition, glass substrates are available with CTE in the range of 3–12 ppm/°C. This allows to mitigate the CTE mismatch between glass and silicon (CTE = 3 ppm/°C) chips and glass and printed wiring board (CTE = 17 ppm/°C).

The ability to structure glass is yet another advantage of having glass core substrates for package and interposer applications. Glass structuring can be of any one of the following types: 1) through-glass vias (TGVs); 2) glass cavities (GCs); or 3) through-glass cavities (TGCs). TGVs can be formed with laser-induced deep etching (LIDE), which first makes local laser modifications on glass, followed by a subsequent wet chemical etching process to minimize the accumulation of microfractures during fabrication. GCs and TGCs can readily be formed with laser machining which may be followed by a wet etching process as necessary. GCs and TGCs are very critical for enabling embedding of dies into the GCs and TGCs in what is known as glass panel embedding (GPE). The cavities of required dimensions are fabricated, and dies are placed into these cavities using automated die pick-and-place tools with few microns level accuracy. GPE process is ideally suited for HI wherein dies with different sizes and functionalities (including passive components such as capacitors and magnetic inductors) are built into the package. In this approach, the capacitors and inductors remain close to where they are needed for applications such as power delivery/IVR. A typical process flow used in GPE is shown in Fig. 6.

With advanced GPE processes, thermal solutions can easily be integrated into the package to remove the heat [101], [102]. For example, in the case of GPE with TGCs, thermal insulation materials and heat spreaders can be attached to the back side of the glass substrate. In the case of GCs, heat can be

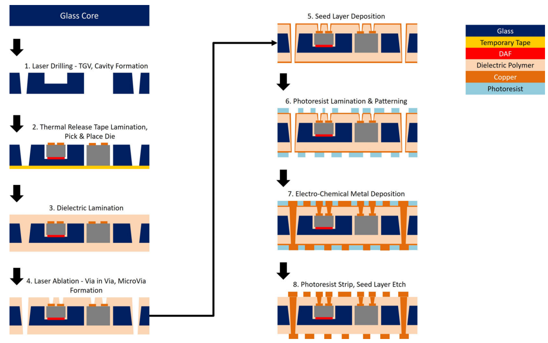

FIGURE 6. **Schematic of a typical GPE process flow.**

removed by incorporating heat sinks after thinning/grinding the substrate. GPE architectures can be easily adapted from a 2.5-D architecture to include 3-D integrations where one of the following approaches can be used: 1) for example, logic dies can be embedded into the glass cavities along with RDLs on both the top and bottom of the glass core following which memory dies can be assembled on the top to generate a 3-D structure with short interconnect distances and much smaller form-factors, thereby reducing the height of the package significantly; 2) passive chips can be embedded into structured glass and multiple dies can be assembled on the glass package structure by the flip-chip process [103]; and 3) furthermore, advanced packaging concepts such as copackaged optics are enabled by GPE wherein one can embed an electronic die into a glass cavity (with thermal solutions described above in place at the backside of the die) and assemble the photonic chip (PIC) on the top of the package. By attaching the PIC on the top, it will be easy to attach a fiber coupler from the top side along with any needed thermal solutions.

Finally, alongside its various superior properties, glass imposes fewer restrictions on substrate format in packaging. While silicon is only capable of being processed in round wafers, glass enables panel processes which reduces the cost. For instance, while a 300-mm wafer can accommodate 2500 packages sized 6 $\times$ 6 mm, a 600 $\times$ 600 mm panel can handle 12 000 packages.

C. CURRENT LIMITATIONS WITH GLASS

The inherent fragility of glass substrates poses a significant challenge, especially as the industry adopts thinner substrates to meet the demands for increased device integration and performance. Thin glass panels, sometimes as thin as 100 μm or less, are particularly vulnerable during handling and manufacturing processes. Such risk of cracking or shattering under stress highlights the necessity for specialized equipment and tailored processes designed to safely handle this material.

In addition to its handling difficulties, glass exhibits relatively low heat dissipation. Although a better thermal conductor than organic laminates, glass is a poor conductor compared with silicon [104]. To overcome the limitations associated with low thermal conductivity of glass, approaches such as incorporating copper structures such as through-package vias (TPVs), copper slugs, and copper traces in RDLs into glass substrates have been demonstrated [105]. In addition, next-generation thermal interface materials (TIMs) for both embedded and substrate-based packages are being actively developed, focusing on reducing the thermal interface resistances to provide maximum heat transfer from the chip.

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